PGR 111, solutions week 04

Task 1 - Concepts & Terminology

Explain the following terminology and database concepts:

- Cartesian product
- (INNER) JOIN
- LEFT (OUTER) JOIN
- RIGHT (OUTER) JOIN

NB: Keep your answers. They might become helpful later in the semester. :-)

Look in the slides, the textbook or on the internet for answers to this one.

Task 2 - SELECT queries

Use the MySQL sample database ("schema") called **world**. (Same as we have been working with earlier.)

Write SQL queries that perform the following:

1. Extract all cities *beginning **and** ending* with 'o' (uppercase first, small at the back) and their population figures. Sort them alphabetically. (Tip: Wildcard. And *no* join in this one.)

   ```
   select Name, Population
   from city
   where Name like 'O%o'
   order by Name asc;
   ```

2. Task 1 gave many city names, but where are these cities located? We want to include the name of the country in which the cities are located. Thus, the columns will be country names, city names and city populations. We want the answer sorted alphabetically by country, then by city.
   Problem: We must query two tables at once! Rewrite the query from task 1 to a JOIN query, where we include the country table, and extract the names of the countries from it.
   Tip: You need to find the common column (foreign key to primary key connection) for these two tables.

   ```
   select country.Name, city.Name, city.Population
   from city
   join country on city.countryCode = country.code
   where city.Name like 'O%o'
   order by country.Name asc, city.Name asc;
   ```

3. Extract all countries (their names). Also have a column showing the names of their capitals (AS Capital), if any. Sort the result by country name. Hint: My answer returns 239 rows - is that the same number of rows that you get? *(Note: The Capital column in country matches the ID column in city.)*

```
SELECT country.Name, city.name
FROM Country
LEFT JOIN City ON country.Capital = city.ID
ORDER BY country.Name;
```

4. Retrieve an overview of all countries (Names) that have at least one city, how many cities they have (AS NumberOfCities) and the average population of these cities (AS AverageCityPopulation).

```
SELECT co.Name, Count(*) AS NumberOfCities,
       AVG(ci.Population) AS AverageCityPopulation
FROM country co JOIN City ci ON co.Code = ci.CountryCode
GROUP BY co.Name;
```

5. Retrieve all the countries in the world (names, as CountryName) where the population of the country is less than 1000 people. We also want to retrieve the population of the country (AS CountryPop) and which continent the country belongs to. In addition, show names of cities (as CityName) in each country if any, and the population of these cities (as CityPop). We want the result sorted alphabetically by CountryName. (Hint: My answer returns 10 rows - is that the same number of rows that you get?)

```
SELECT country.Name AS CountryName,
       country.Population AS CountryPop,
       country.Continent,
       city.Name AS CityName,
       city.Population AS CityPop
FROM country
LEFT JOIN city ON Code = CountryCode
WHERE country.Population < 1000
ORDER BY country.Name ASC;
```

6. List the cities of the world located in a country with some form of monarchical system of government. Show the city's name, the city's population, and the country's government form. Sort by the city with the highest population first.

```
SELECT ci.Name, ci.Population, co.GovernmentForm
FROM Country co
JOIN City ci ON ci.CountryCode = co.Code
WHERE co.GovernmentForm LIKE '%Monarchy%'
ORDER BY ci.Population DESC;
```

7. Retrieve a list of cities (their names and population) in the world that have at least 8,000,000 inhabitants, and which country and continent they belong to. Sort by the cities' population, largest first. Use aliases (AS) if there are ambiguous column names in the result.

```
SELECT ci.Name AS CityName, ci.Population,
       co.Name AS CountryName, co.Continent
FROM Country AS co
JOIN City AS ci ON co.Code = ci.CountryCode
WHERE ci.Population >= 8000000
ORDER BY ci.Population DESC;
```

8. Create a query that extracts an overview of the countries in Asia where at least 10 different languages are spoken. The query should contain: Name of country and number of languages. It should be sorted descending by number of languages.

```
SELECT Name, count(*) AS NumberOfLanguages
FROM countrylanguage
JOIN country ON CountryCode = Code
WHERE Continent = 'Asia'
GROUP BY Name
HAVING NumberOfLanguages > 9  -- Or: >= 10
ORDER BY NumberOfLanguages DESC;
```

Task 3 – Create your own SELECTs

Regarding the world database, come up with some interesting queries you want to execute. Feel free to use different JOINs, GROUP BY, composite WHERE-clauses (AND/OR), HAVING and using wildcards! :-D

Once you have created some tasks and posted them on the Canvas forum, solve tasks from others with a level of difficulty that suits you. :-)